Self-Supervised Learning on MRI Data

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1 Aims

Lesion segmentation plays a critical role in a variety clinical tasks such as medical diagnosis, prognosis, and treatment. In most cases, segmentation is performed manually by the radiologist, making the process very time-consuming and dependent on understanding of the human anatomy and expertise. Recent advances in deep learning, and generative modeling in particular, have proven to be promising tools for detecting arbitrary anomalies in data and automating medical analysis, relieving professionals from time-consuming tasks [1]. However, training supervised models can often be a significant challenge. The reason for this is that training neural networks typically requires large amounts of labeled data, which is often difficult to obtain due to disease heterogeneity and patient privacy aspects. Therefore, we need to consider such machine learning models that do not require vast amounts of labeled data and can properly generalize to rare diseases that may not have been seen during training. One of the possible approaches is to use models based on variational inference to model the healthy distribution of the human brain in a low-dimensional space and then, during inference, detect anomalous data passed as input to that model as outlier. This approach is more similar to how radiologists make their decisions, as it does not require manual annotation and relies only on a priori knowledge of pathologies manifestations. [2]. Although there are several possible ways to address this task, our project will focus mainly on probabilistic graphical models such as Variational Autoencoder (VAE) [3] and it's extensions [4] [5]. VAEs learn the distribution of the data by mapping it to the normative distribution in the latent space and do not require any labels for learning. Once trained, the model will produce different latent encodings for the unhealthy samples as they will have a different distribution than the one it has learned. Their reconstructions can then be segmented via various morphological operators or MAP-based methods [6] and compared to the ground truth segmentations.

2 Objectives

2.1 Primary Objectives

1. Investigate and compare some of the available medical image analysis methods for lesion detection and segmentation.
2. Implement the described architectures in Python using PyTorch deep learning framework.

3. Examine the performance of Autoencoders, Variational Autoencoders in relation to lesion detection and segmentation tasks.
4. Research the effects of domain shift and the ability of models to generalize to new data with different distributions.

2.2 Extension Objectives

1. Explore the extensions of VAEs and compare their results to classic VAE and AE results.
2. Study the weaknesses of the demonstrated methods and discuss potential improvements.
3. Examine the amount of data needed for proper model generalization and inference.
4. Study how the size of the latent space affects the quality of segmentation.

3 Dataset

Our project will use T1-weighted images from the IXI Dataset[1] to model the healthy distribution of the data in the latent space. Then, BraTS2020 dataset [7–9] which contains pre-operative scans of glioblastoma and lower grade glioma will be used for anomaly detection and segmentation tasks. In addition, we will use ATLAS[2] dataset to test the ability of the model to generalize to new domains. Both 'unhealthy' datasets will use T-1 weighted images, similar to the data used to train the model.

4 Relevance

Generative modeling has already shown impressive results in anomaly detection, but machines are still not as good at this task as humans. Since the number of experts who can detect the anomaly does not grow as far as the number of patients to be studied [10], we need to explore new approaches to automate this process. Our task is to explore some of the state-of-the-art methods and evaluate how good they are for a lesion segmentation task. As mentioned, this approach is more human-like than supervised deep learning, and, in a broader perspective, could give us a more reliable and faster way to detect and treat certain diseases at early stages.

[1] https://brain-development.org/ixi-dataset/
[2] https://fcon_1000.projects.nitrc.org/indi/retro/atlas.html.

5 Project Timescale

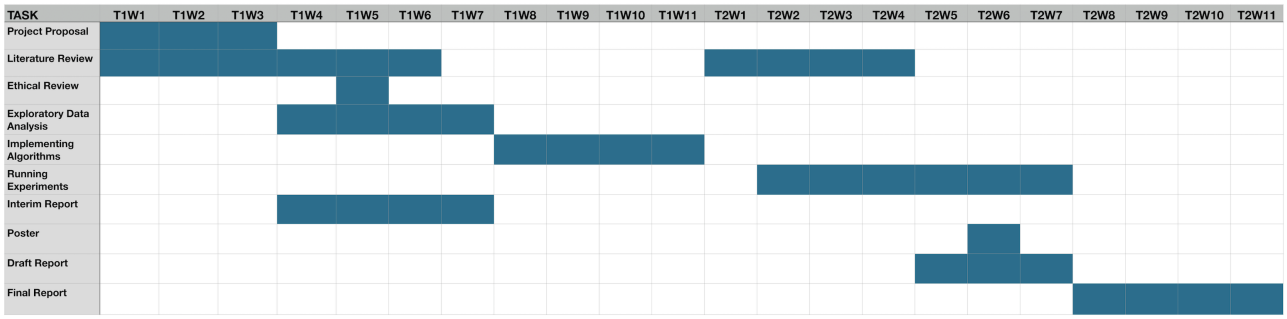

Figure 1: Project timescale.

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